

Dataset & Setup	
Records analysed	25 (20)
Channel configuration	4-channel (13-unipolar → 5-bipolar)
Native sampling rate	1000 Hz (400 Hz)
Segment length analysed	60 s (60 s)

Stage 1 — Baseline Wander Removal	
Cutoff frequency	3 Hz (3 Hz)
Window function	Hamming (not in paper)
FIR taps ($\text{round}(1000\text{-fs}/400)$)	2501 (1000)

Stage 2 — Power-Line Interference Cancellation	
Mains frequency	50 Hz (50 Hz)
Harmonics cancelled	4, up to 200 Hz (not in paper)
PLL gain tuning (τ , ζ , $\omega_{\text{r}}/\omega_{\text{carrier}}$)	0.13 s, 1.0, 0.04 (not in paper)
Isolation filter bandwidth	± 10 Hz (not in paper)
QRS-blocking (threshold, dilation)	$\sqrt{2}\cdot\sigma$, ± 50 ms (not in paper)
Warm-start / max freq. drift	2 s, ± 4 Hz (not in paper)

Stage 3 — Upsampling	
Resample rate	1000 → 2000 Hz (400 → 2000 Hz)

Stage 4 — Maternal QRS Detection & MECG Cancellation	
QRS window	± 0.05 s (± 0.05 s)
P-wave window	0.20 s (0.20 s)
T-wave window	0.40 s (0.40 s)
Template history (N beats)	10 (10)
PCA components (enhancement)	1 (1)
Template robustness	min/max p-p trimming (not in paper)
Least-squares scaling	Ridge $\lambda=1\text{e-}3$ (unregularised)
Detection thresholds / refractory	2σ / 1.5σ , 0.35 s (not in paper)

Stage 5 — Fetal QRS Extraction	
Rough-candidate template width	0.05 s (not in paper)
Refractory period (3.3 Hz tachycardia bound)	0.3 s (0.3 s)
Detection thresholds	2σ / 1.5σ (not in paper)
Beats averaged (N_{av})	150 (150)
Averaging window duration	0.25 s / 500 samples (not in paper)
SNR gain from averaging	$\sqrt{150} \approx 12.2\times$ ($\sqrt{150} = 12$)

ICA Baseline	
Algorithm	FastICA (JADE)
Input channels	4 (13 unipolar)
Component count	n_channels (n_channels)
Whitening	unit-variance (not in paper)
Random seed	42 (not in paper)
Best-component selection	automatic, max valid-FHR count (visual inspection)

Evaluation Metrics & Thresholds	
Success / coverage threshold	60% duration-weighted (not in paper — qualitative criterion)
Reliability outlier rule	> 10 bpm from 10 s-block median (identical)
Ground-truth match tolerance	± 50 ms (not in paper)
Tolerance sensitivity sweep	10/25/50/75/100 ms (not in paper)
FHR physiological bounds	78–198 bpm (78–198 bpm)
FHR MAE/RMSE grid step	1 s (not in paper)
Spearman bootstrap	n=2000, 95% CI (not in paper)